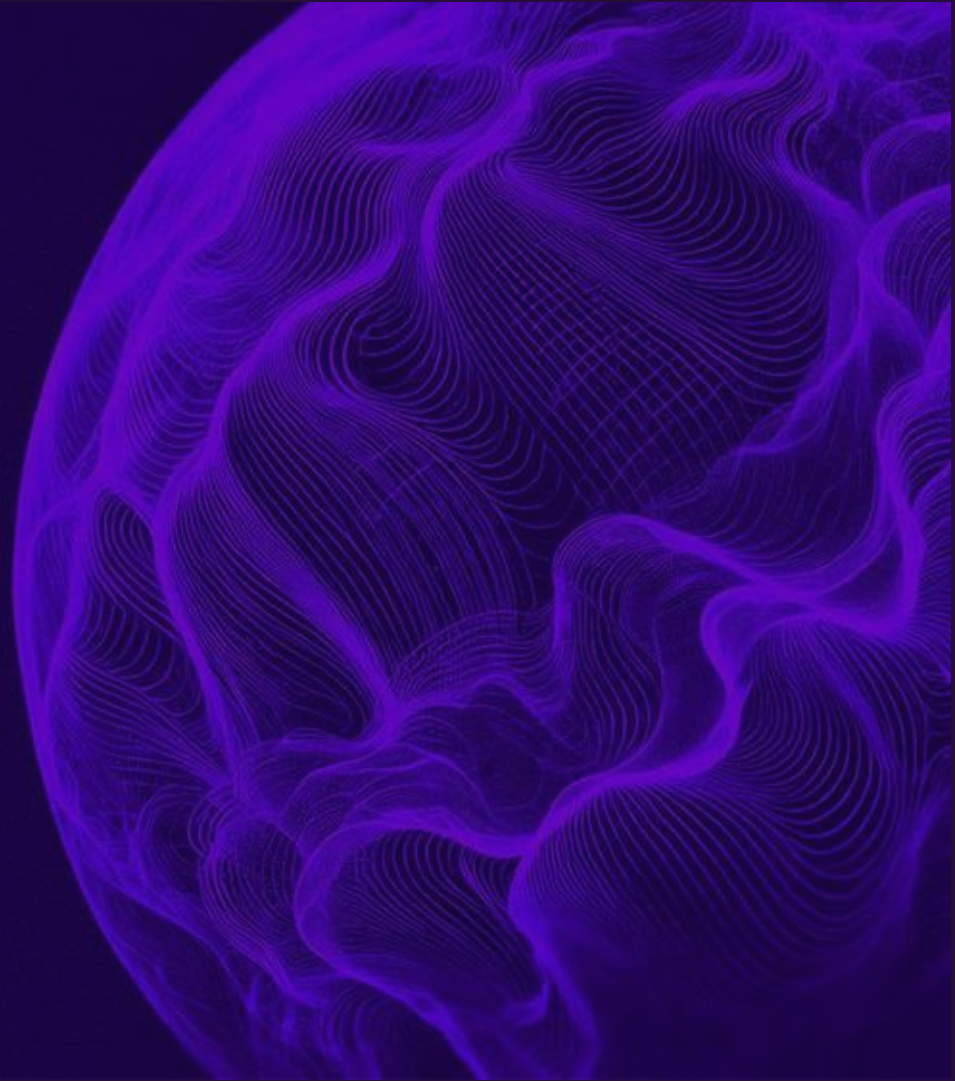


Accelerating AI with SambaNova

Timothy Clarke
Director Public Sector Accounts

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The SambaNova Journey

Snapshot:

- Founded in 2017 by AI pioneers with roots at Stanford University
- 400+ employees across Palo Alto, Austin, London, Bangalore, Tokyo, Singapore, and Australia
- $1.4B+ funding raised through Series E
- $1B Series F close at $11B post-money valuation

Founders



Rodrigo Liang
Co-founder & CEO



Kunle Olukotun
Co-founder & Chief Technologist & Stanford Professor



Christopher Ré
Co-founder & Stanford Professor



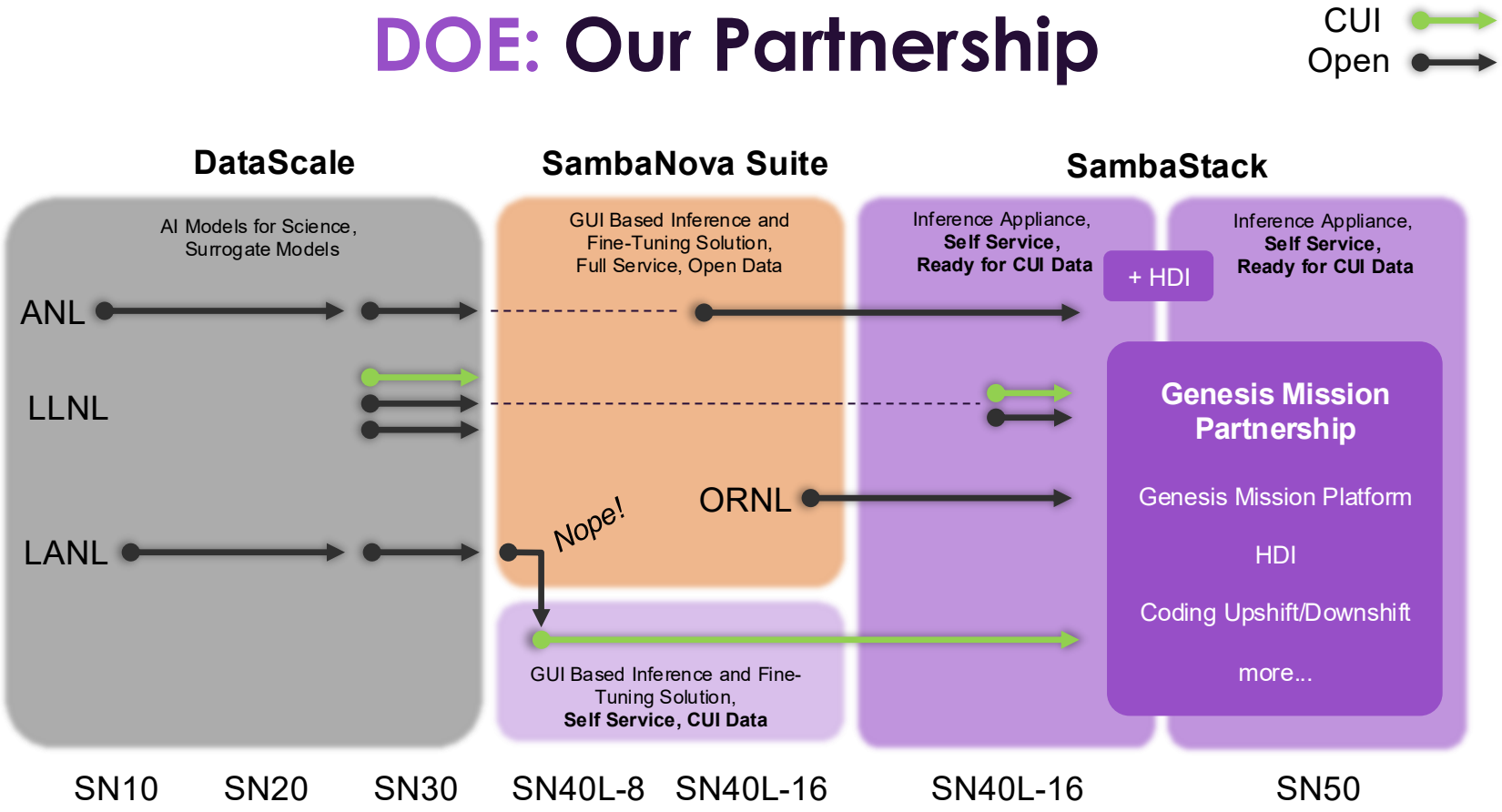
Lip-Bu Tan
Chairman

Sophisticated, long-term investors and strategic new investors

General Atlantic · Vista · intel capital · BlackRock · Seligman Ventures · T.Rowe Price
Capital Group · Cambium Capital · GV · SoftBank · Temasek · QIA
Battery · Samsung Catalyst Fund · GIC · Walden International · Mayfield · Celesta

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DOE: Our Partnership



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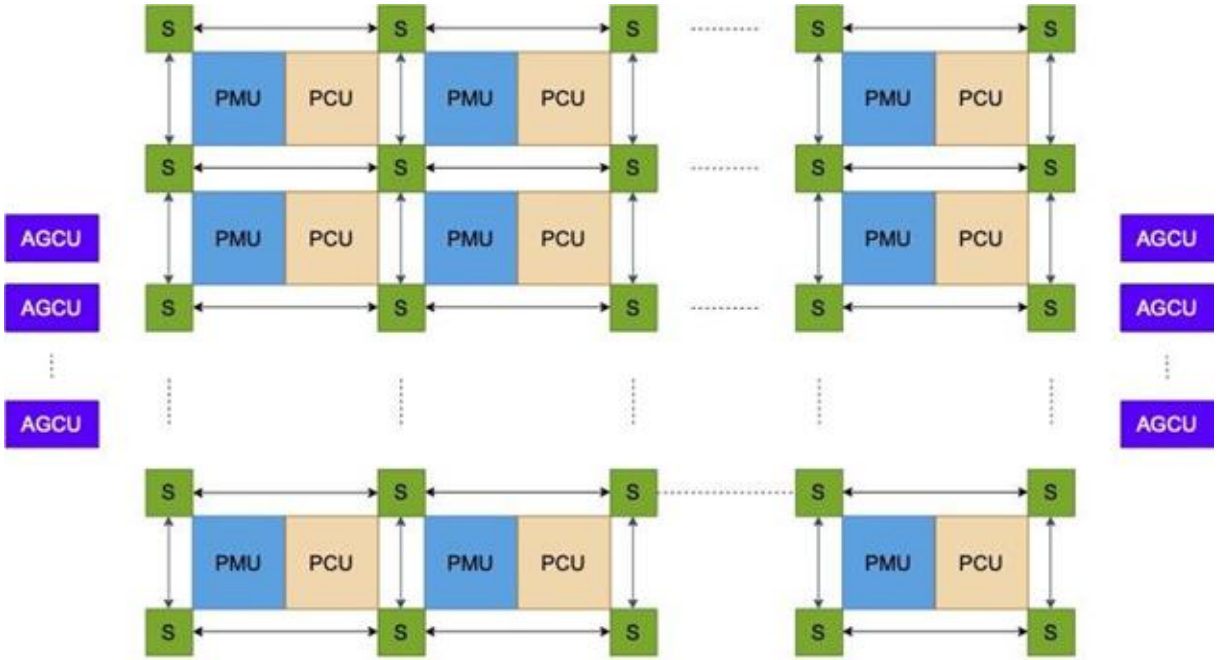
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TACC 25 YEARS OF POWERING DISCOVERY



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RDU: Tile Architecture



PCU: Compute unit

PMU: Memory unit

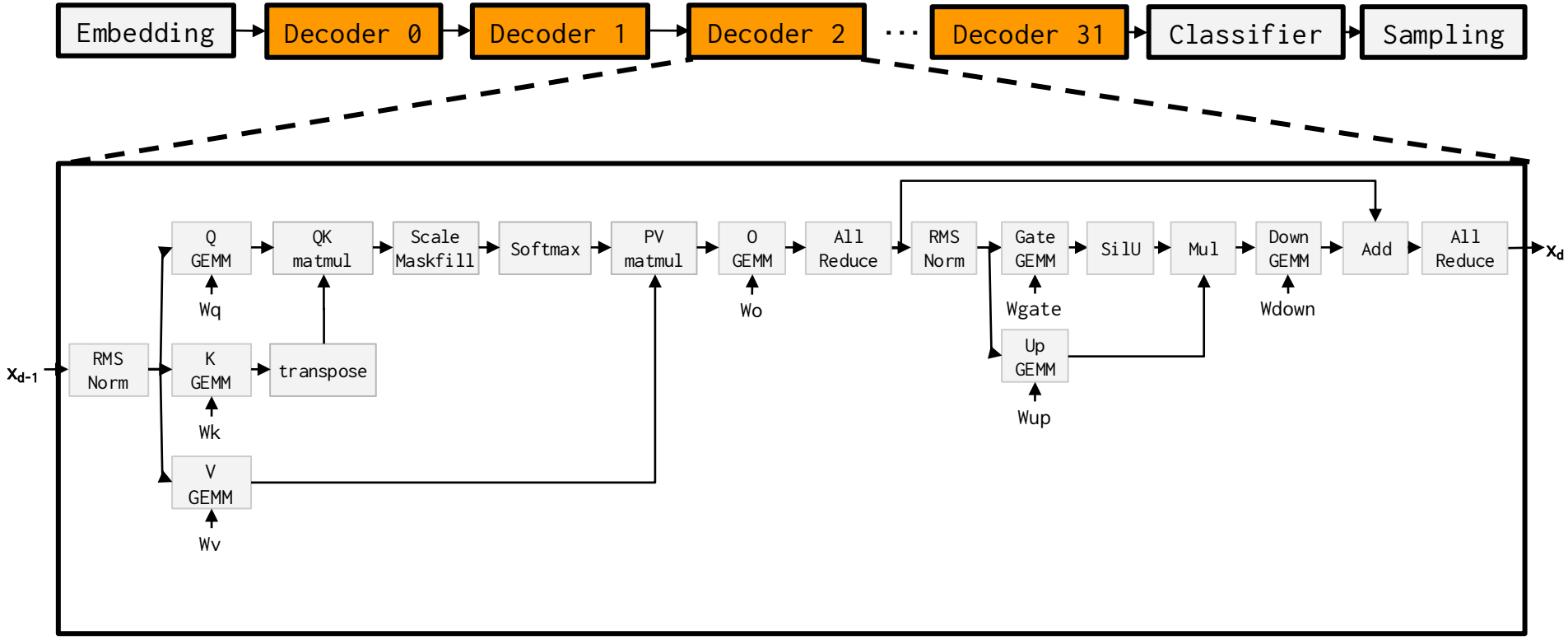
S: Mesh switches

AGCU: Portal to off-chip memory and IO

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Structure of a Transformer

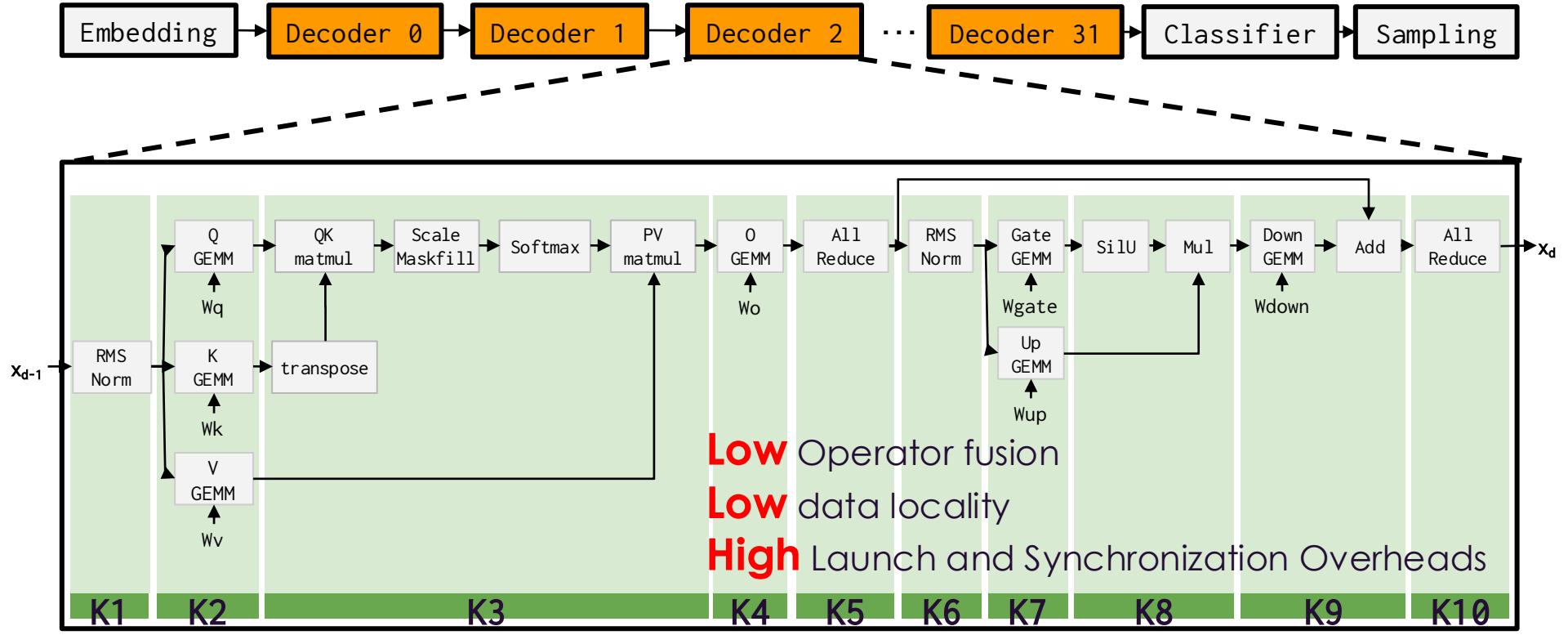
Example: Llama3.1 8B



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GPUs Incur Overhead with Low Operator Fusion

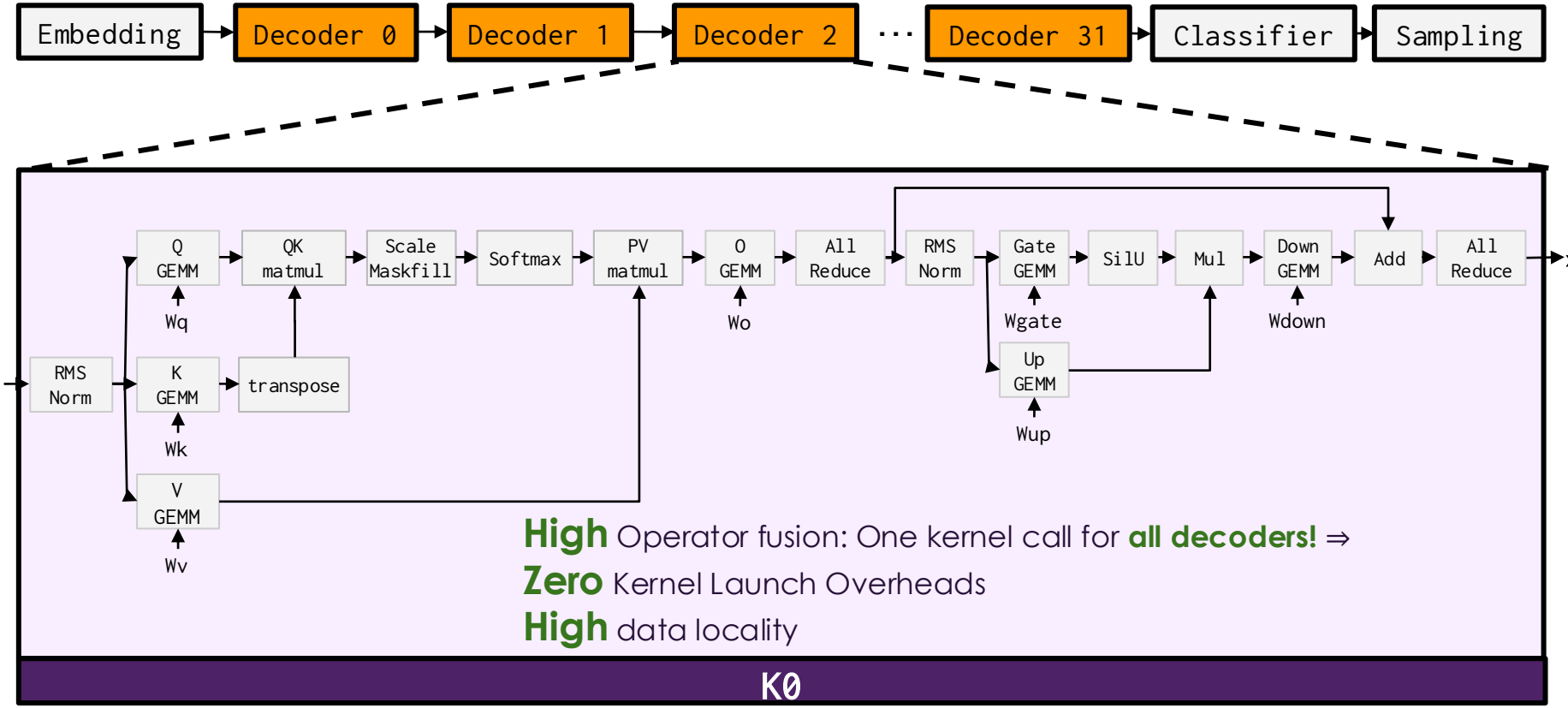
Example: Llama3.1 8B



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SN40L RDU Fuses the Entire Decoder into One Kernel!

Example: Llama3.1 8B



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SambaNova SN50 Snapshot

Hardware Features

Speed AND Throughput

Dataflow Architecture

Agentic Caching

Cloud Scale

5X Compute
Vs SN40

4X Network BW
Vs SN40

5X Max Speed
Vs Nvidia Blackwell

3X Throughput
For Agentic Inference



Solving AI's Data Movement Problem

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Agentic inferencing can be slow & expensive. Matching the right chip to the workload is the solution.

Each stage of an AI agent's inference loop has a different hardware winner.

Great Fit on GPU

Stage 01 • Prefill

Understanding the prompt

Parallel processing of the full input context — compute-intensive and batch-friendly.

Best fit, data movement optimized

Stage 02 • Decode

Answering the prompt

Token-by-token generation — memory bandwidth-bound, latency-critical for agentic loops.

Ideal fit for CPUs

Stage 03 • Action

Executing functions & tool calls

API calls, code execution, vector DB lookups — serial, branching, OS-level tasks.

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Agenda



1 Who We Are

2 How it Works

3 SN50 / HDI

4 The Fun Stuff

We Made It!

LLM-Assisted Translation of Legacy FORTRAN

Currently working on a study with Aran Biswas (LANL) to extend their research into methods for LLM-Assisted Translation

Presenting my initial findings at Supercomputing26 on Monday

- On-Prem vs. Cloud tokens / wall clock
- Auto Refactoring
- Efficiency Directives

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arXiv:2504.15424v1 [cs.SE] 21 Apr 2025

LLM-Assisted Translation of Legacy FORTRAN Codes to C++: A Cross-Platform Study

Nishath Rajiv Ranasinghe[1], Shawn M. Jones[1], Michal Kucer[1], Ayan Biswas[1], Daniel O'Malley[1], Alexander Buschmann Most[1], Selma Liliane Wanna[1], Ajay Sreekumar[2]

[1]Los Alamos National Laboratory, Los Alamos NM 87545,
[2]School of Information, University of Arizona, 103 E 2nd St #4, Tucson, AZ 85721

Abstract

Large Language Models (LLMs) are increasingly being leveraged for generating and translating scientific computer codes by both domain-experts and non-domain experts. Fortran has served as one of the go to programming languages in legacy high-performance computing (HPC) for scientific discoveries. Despite growing adoption, LLM-based code translation of legacy code-bases has not been thoroughly assessed or quantified for its usability. Here, we studied the applicability of LLM-based translation of Fortran to C++ as a step towards building an agentic-workflow using open-weight LLMs on two different computational platforms. We statistically quantified the compilation accuracy of the translated C++ codes, measured the similarity of the LLM translated code to the human translated C++ code, and statistically quantified the output similarity of the Fortran to C++ translation.

1 Introduction

A Large volume of scientific computational software implemented in HPC environments has been written in programming languages such as Fortran and C due to their superior performance. However, recent advancements in computer hardware are not fully utilized by older generations of Fortran, and these legacy codes often encounter difficulties with memory allocations. There is a lack of human resources to maintain and improve these code-bases for mission critical applications in the future (Shipman and Randles, 2023; Pietrini et al., 2024).

agentic workflows, there is a need for systematic methods to evaluate the effectiveness of machine generated scientific software.

However, very few studies exist for LLM-assisted code translation from Fortran to C++, primarily due to a lack of quality training data sets. A recent study (Lei et al., 2023), compiled pairs of OpenMP Fortran and equivalent C++ codes to evaluate LLM code translation and evaluated their results using both quantitative (e.g., CodeBLEU score (Ren et al., 2020)) and qualitative approaches (e.g., human evaluation). There is also a lack of LLM-based Fortran to C++ code translation tools that can be readily deployed to assist developers in mission critical and secure environments. Furthermore, earlier attempts to translate code from Fortran to C++ have not accounted for successful compiles or output evaluation of the translated code (Theurich et al., 2001).

In this study, we make several contributions. We conduct an analysis of translating open-source code-bases using open-weight models. Our workflow (Figure 1) is designed to be agnostic of any specific LLM or computational platform (e.g., vLLM), building towards a set of standardized evaluation measures for machine-generated code translation. We evaluate the similarity to the human-translated target code using the common CodeBLEU measure (Ren et al., 2020), how much of the translated code compiles (compilation accuracy (Wen et al., 2022a)), and how well the output of



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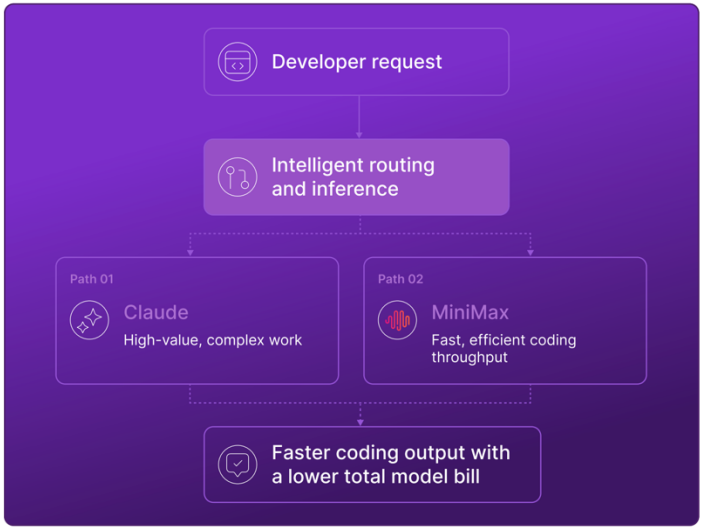
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Automatic Model Upshift / Downshift

THE SAMBANOVA APPROACH

Route Each Coding Task To The Right Model

A blended workflow keeps high-value frontier models in the stack while routing suitable coding tasks to fast, efficient open-source models running on SambaNova.



Outcomes

01

50% lower model bill

Use a smarter model mix to improve workflow economics.

02

3x faster coding output

Increase development velocity with faster model responses.

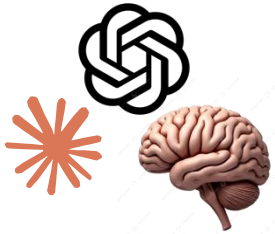
03

Preferred models remain

Claude stays in the workflow for tasks where frontier-model capabilities matter most.

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Developer Request



Intelligent Routing

LiteLLM

openrouter

portkey

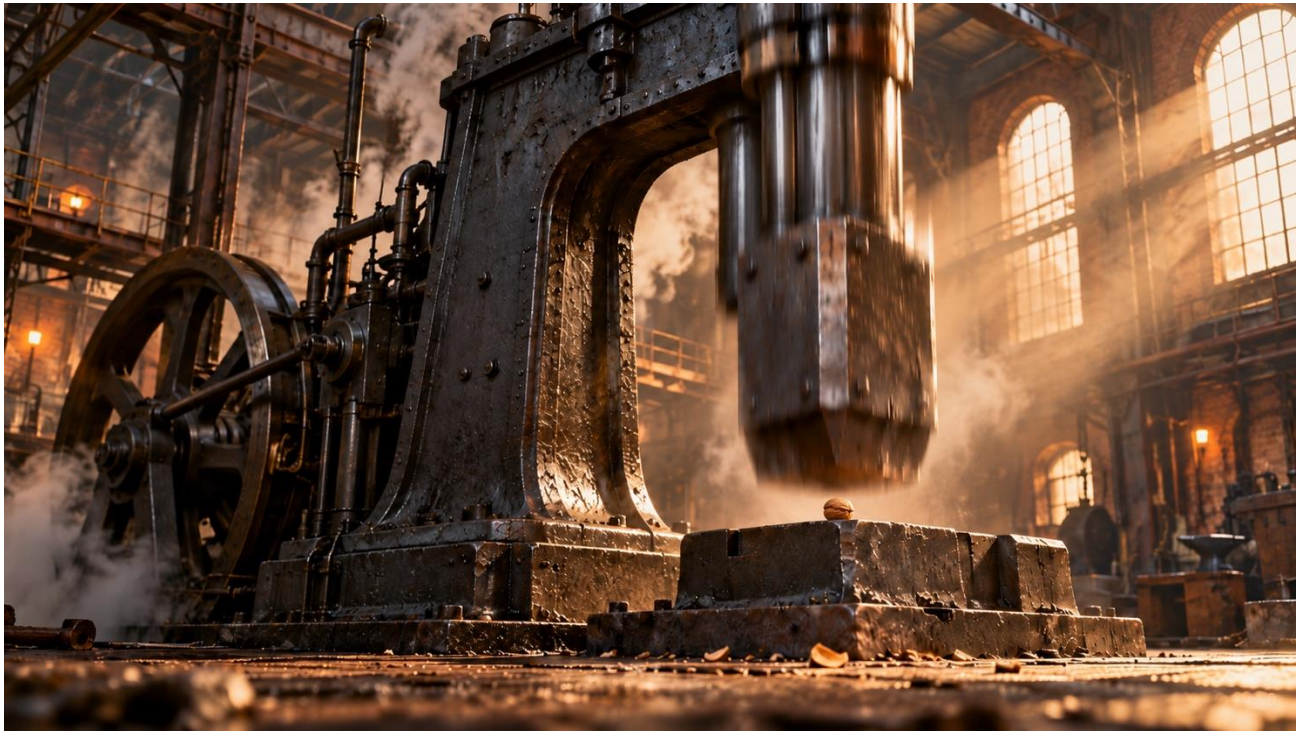
A PALO ALTO NETWORKS COMPANY

Cloud-based Foundation Model

On-premises Open Weight Model

Saving Lots of Money

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"You must not use a steam hammer to crack a nut, if a nutcracker would do."

— Lord Diplock, British Law Lord (R v Goldstein, 1983)

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Implementation Study

Task: Fortran 90 → C++ Translation (John Burkardt)

Foundation Model Creates Implementation and Smaller Models do tasks on-premises



Method 1

Foundation Model Orchestrates smaller Models doing tasks on-premises



Method 2

Foundation Model does every task

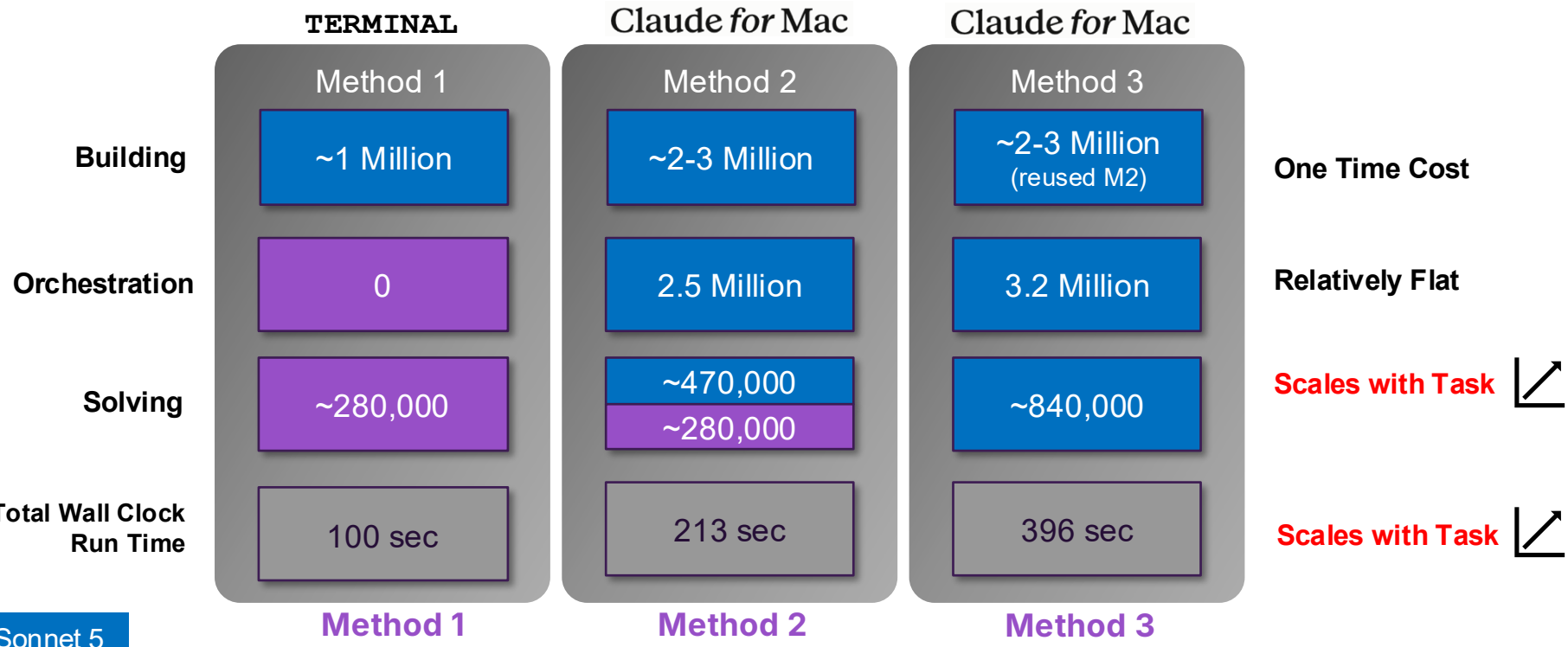


Method 3

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Token Consumption Study

Task: Fortran 90 → C++ Translation (John Burkardt)



Sonnet 5

Gemma 4

Translate **20 examples** of John Burkardt's Fortran 90 in C++ that compiles only (QA check later). Utilize 8-way Parallelism for speedup.

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Model Translation Quality Study

Task: Fortran 90 → C++ Translation (John Burkardt)

Claude - Sonnet 5

Does The C++ Function Correctly?

55%

Round 1

Choice 1: Work on 100% of the Tasks and have 55% Success

34 minutes

SN40 - Gemma 4

Does The C++ Function Correctly?

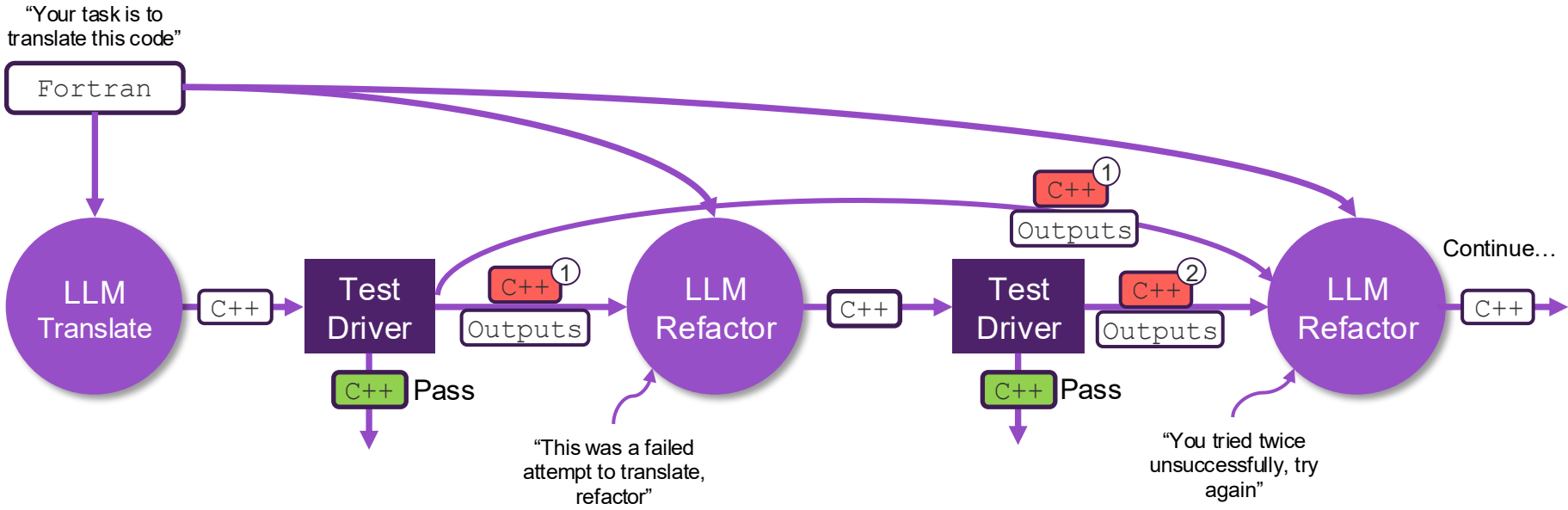
26%

Round 1

Choice 2: Work on 100% of the Tasks and have 26% Success

14 minutes

Exact Task: QA Check on **200 examples** of John Burkardt's Fortran 90 in C++

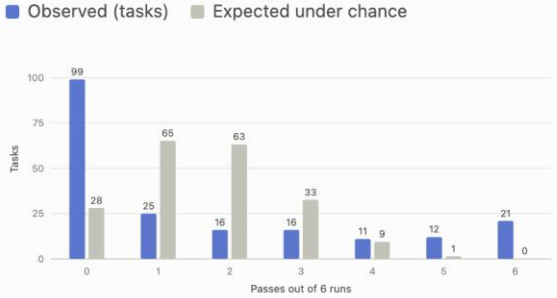


Outputs:
compile errors,
runtime errors,
numeric diff,
reference values

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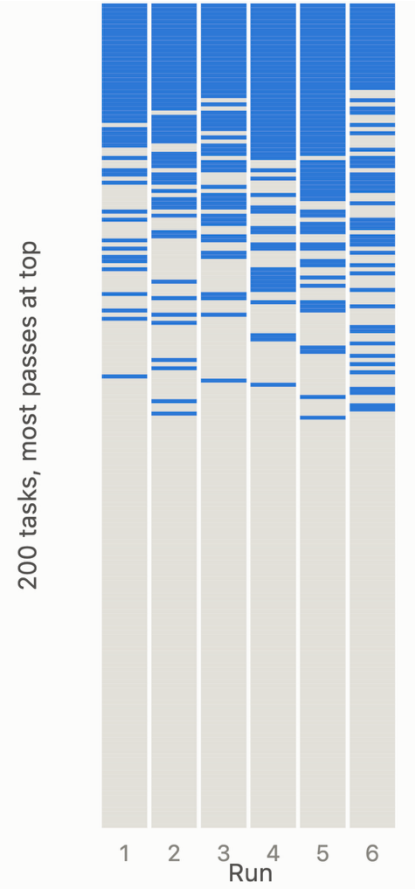
Observed vs. random chance

Tasks by number of passes out of 6. Random chance = binomial expectation at the pooled rate.



▼ Table view

Passes	Observed	Expected
0	99	28.1
1	25	65.2
2	16	63.1
3	16	32.6
4	11	9.5
5	12	1.5
6	21	0.1

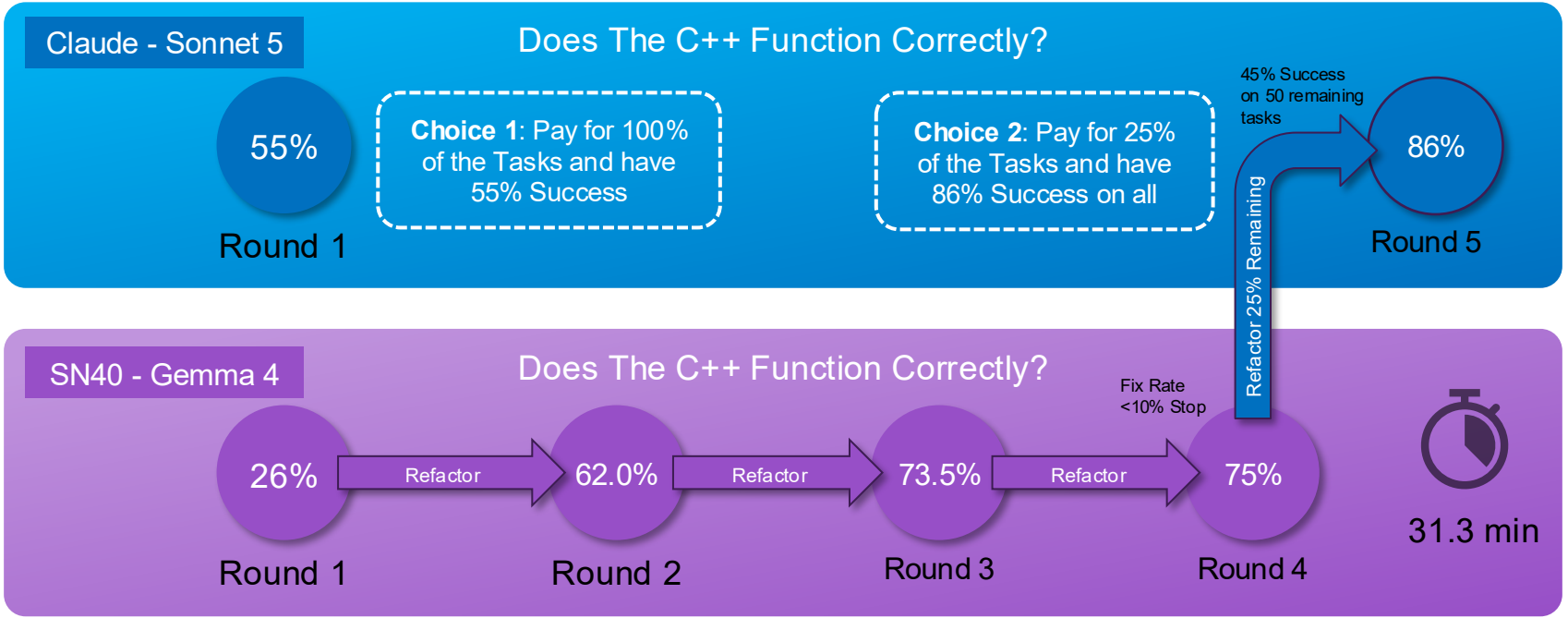


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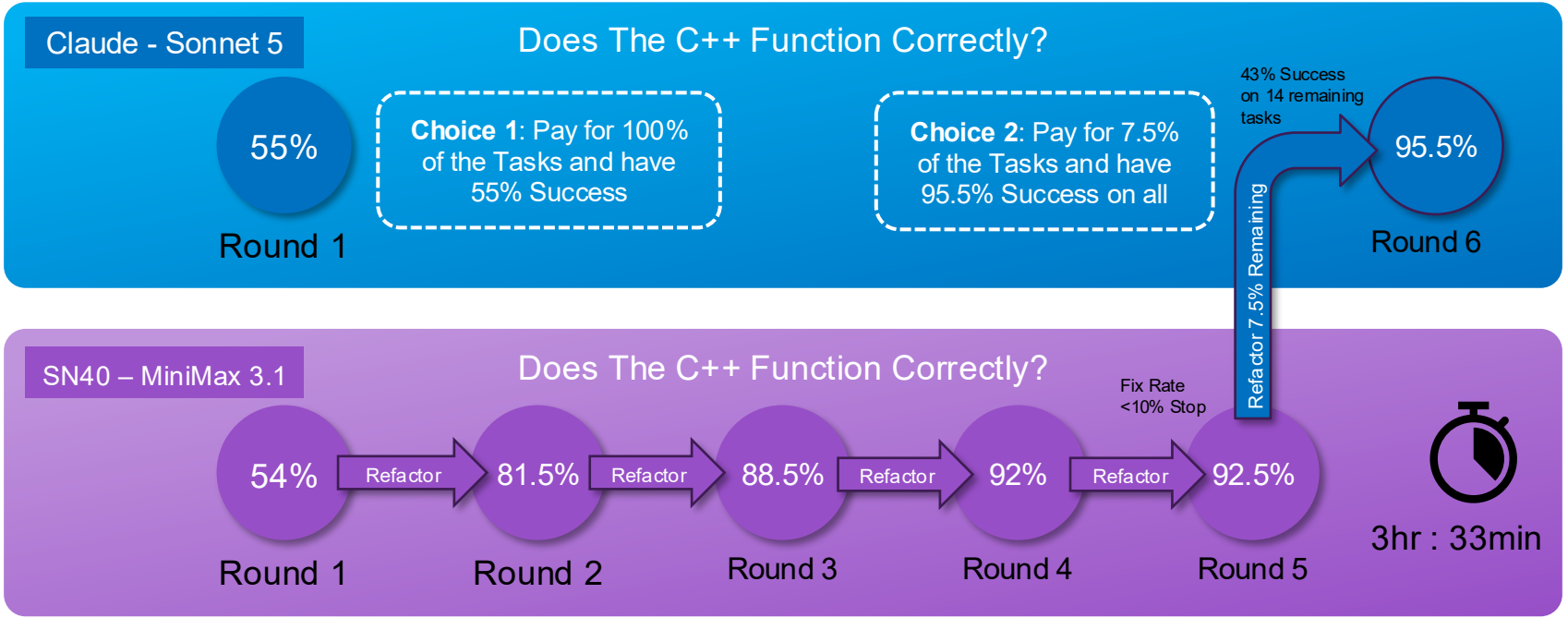
Model Translation Quality Study (Gemma4)

Task: FORTRAN 90 → C++ Translation (John Burkardt)



Exact Task: QA Check on **200 examples** of John Burkardt's Fortran 90 in C++

Model Translation Quality Study (MiniMax 3.1)
Task: FORTRAN 90 → C++ Translation (John Burkardt)



Exact Task: QA Check on **200 examples** of John Burkardt's Fortran 90 in C++

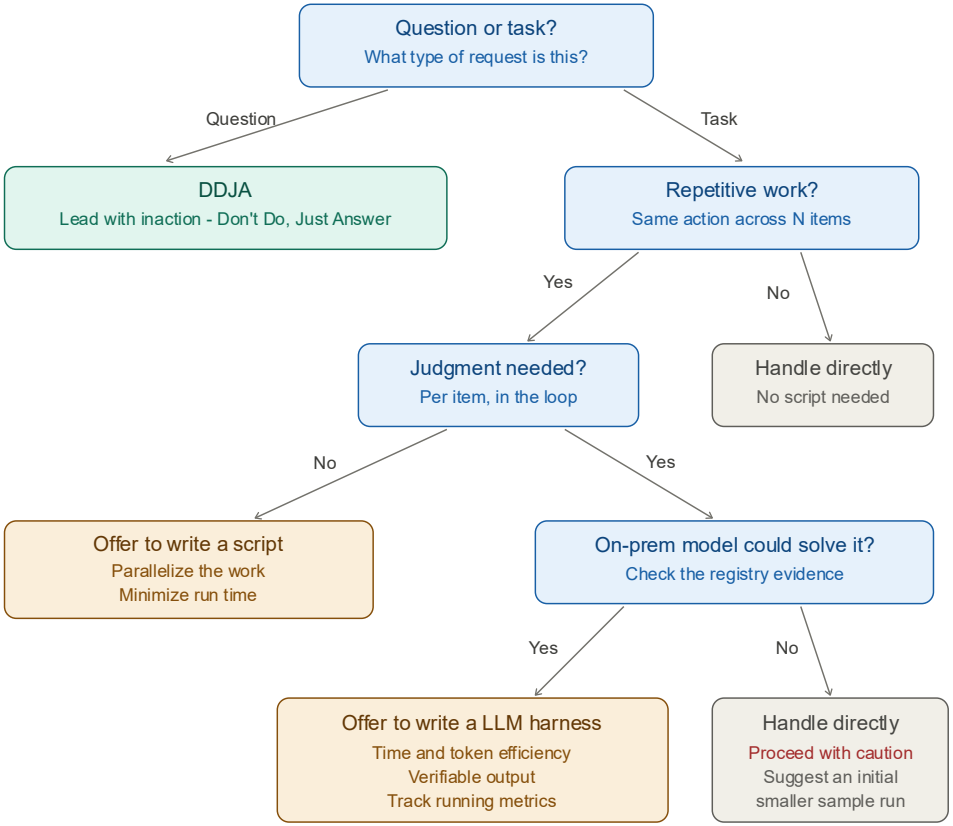
Coding: Model Upshift / Downshift



Automatic or Manual – Which is better for token spend?

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Efficiency Directives



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Time for Demo!

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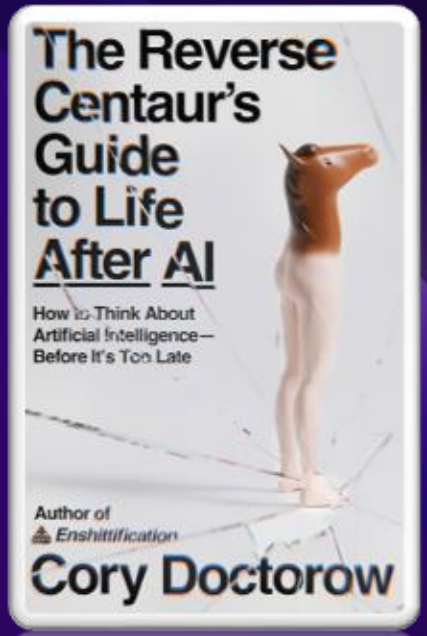
THANK YOU

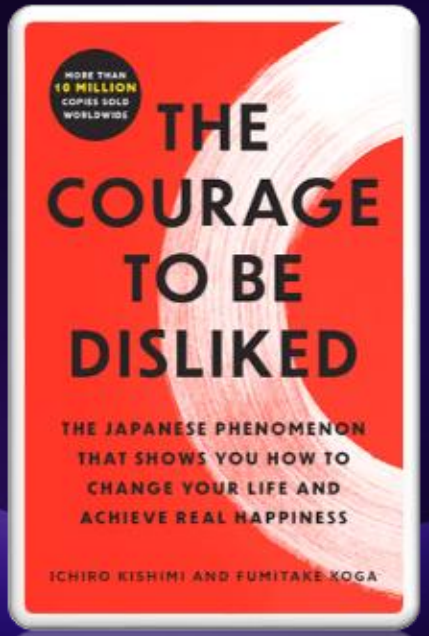
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What I am reading…





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